

News Pulse - AWS Deployment & Troubleshooting Guide

Goal

Deploy a production-ready Node.js backend on AWS EC2, migrate PostgreSQL data to AWS RDS, keep the backend alive with PM2, expose it through Nginx, and verify public access.

Architecture

Local PC → GitHub → EC2 (Ubuntu) → Node.js Backend (PM2) → Nginx → AWS RDS
PostgreSQL → Browser

Deployment Steps

1. Launch EC2 and SSH with the .pem key.
2. Clone GitHub repository.
3. Create RDS PostgreSQL.
4. Configure RDS Security Groups.
5. Export local DB using pg_dump.
6. Copy dump with scp.
7. Restore with pg_restore.
8. Configure DATABASE_URL.
9. Run backend with PM2.
10. Configure Nginx reverse proxy.

Useful Commands

SSH: `ssh -i key.pem ubuntu@IP`

Clone: `git clone REPO_URL`

Export: `pg_dump -Fc -f news_pulse.dump news_pulse`

Copy: `scp news_pulse.dump ubuntu@IP:/home/ubuntu/`

Restore: `pg_restore -h RDS -U postgres -d news_pulse --clean --no-owner dumpfile`

Verify: `\dt, SELECT COUNT(*) FROM articles;, SELECT COUNT(*) FROM clusters;`

PM2: `pm2 start, pm2 save, pm2 startup`

Nginx: `sudo nginx -t, sudo systemctl restart nginx`

Errors & Why They Happened

Bad .pem permissions: Windows security permissions.

psql not found: PostgreSQL client missing.

Connection timeout: Security Group/network issue.

Password authentication failed: Wrong RDS password.

'relation does not exist' during restore: Expected with --clean on an empty DB.

self-signed certificate in certificate chain: Node pg SSL verification.

getaddrinfo EAI_AGAIN: Malformed DATABASE_URL in .env (DATABASE_URL duplicated).

Cannot GET /: Root route not implemented; use /health.

How We Fixed Them

Adjusted .pem permissions, installed PostgreSQL client, opened port 5432 between EC2 and RDS, corrected credentials, verified restore with SQL counts, configured pg SSL (rejectUnauthorized:false), corrected DATABASE_URL, restarted PM2 with --update-env, configured Nginx.

SQL Verification

\dt lists tables.

SELECT COUNT(*) FROM articles; verified 140 articles.

SELECT COUNT(*) FROM clusters; verified 36 clusters.

Lessons

Always verify networking before code, read error messages carefully, test RDS with psql first, verify migrations after restore, restart PM2 with --update-env after .env changes, and use /health endpoints to validate deployments.

Outcome

Production backend successfully running on EC2, connected to AWS RDS, managed by PM2, served via Nginx, and publicly reachable.